TEMCO LEARN

Register maps

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Register maps

The address contract: the TStat Modbus register map, the BACnet points each T3 model publishes, and the registers that configure a WireGuard tunnel.

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Reading a register map

A register map is the contract between a controller and whatever is talking to it. These pages are the maps themselves: the address, the width, and what the value means.

Map	What it is for
TStat registers	The Modbus map of the TStat, in address order
BACnet points by controller	The object names each T3 model publishes, and how many
WireGuard registers	The registers that configure a WireGuard tunnel on a T3 ESP32 controller

The TStat map is the longest and the one support asks about most: a reading that looks wrong is usually a register read at the wrong width.

TStat registers, part 1

The Modbus register map for the TStat, in address order. Addresses are decimal, and where the map gives a range the register occupies every address in it. This part begins at address 0 to 3 and ends at 208.0.

Address	Bytes	Register and description
0 to 3	4.0	Serial Number - 4 byte value. Read-only
4 to 5	2.0	Software Version – 2 byte value. Read-only
6.0	1.0	ADDRESS. Modbus device address
7.0	1.0	Product Model. This is a read-only register that is used by the microcontroller to determine the product
8.0	1.0	Hardware Revision. This is a read-only register that is used by the microcontroller to determine the hardware rev
9.0	1.0	PIC firmware version
10.0	1.0	PLUG_N_PLAY_ADDRESS, 'plug n play' address, used by the network master to resolve address conflicts. See VC code for algorithms
15.0	1.0	Base address selection.0 = Protocol address,1 = PLC address.
16.0	1.0	Firmware Update Register, used to show the status of firmware updates.Writing 143 to register 16 - resets the unit to latest factory defaults.Writing 159 to register 16 - makes the current configuration the new factory default.
17,16,19		Spare
20.0	2.0	Hardware Options Register, starting with LSB: Bit0=Clock present or not, Bit1 = Humidity present or not, Bit2 = C02 Sensor, Bit3=CO sensor, Bit4 = Motion Sensor
21 to 100		Blank, for future use
101.0	2.0	TEMPERATURE reading in DegC or F from the sensor used in the control loop PI1 which is configured in register 111. This can be the internal sensor, external, or an average of the two. Writing a temperature value to this register will calibrate the currently selected sensor by adjusting the associated calibration register. If average is selected then you cannot write to this register.
102.0	2.0	COOLING_VALVE, a number from 0-1000 representing 0% (closed) to 100% (open). When Range = On/Off mode, On=1000, Off=0.
103.0	2.0	HEATING_VALVE, a number from 0-1000 representing 0% (closed) to 100% (open)
104.0	2.0	PID1, current PI calculation for PID number 1 (PID2 is in register 240)
105.0		NOT USED FOR REV 25
106.0	1.0	COOL_HEAT_MODE, heating or cooling mode. 0=none, 1=cooling, 2=heating.

Address	Bytes	Register and description
107.0	1.0	MODE_OPERATION, heating or cooling state: 0-7 = coasting, cooling 1,2,3, heating 1,2,3
108.0	1.0	DIGITAL_OUTPUT_STATE, bit 0 thru 4 = relay 1 thru 5.
109.0	2.0	CALIBRATION, this is the calibration factor for the internal sensor, normally maintained by the tstat.
110.0	2.0	CALIBRATION_EXTERNAL , this is the calibration factor for the external sensor, normally Maintained by the tstat.
111.0	1.0	TEMP_SELECT, Sensor to be used for the PID calculations, 1= external sensor analog input 1 , 2 = internal thermistor, 3 = average the internal thermistor and analog input1
112.0	1.0	DAC_OFFSET , Calibration data for the 0-10VDC signal, internal variable maintained by tstat
113.0	1.0	NOT USED FOR REV 25
114.0	1.0	PTERM , proportional term for PI calculation
115.0	1.0	ITERM , integral term for PI calculation
116.0	1.0	NOT USED FOR REV 25
117.0	1.0	NOT USED FOR REV 25
118.0	1.0	SEQUENCE , 0 = internal test sequence, ooutputs slowly cycle on/off or ramp up & down. 1 = normal, operation according to the output tables.
119.0	1.0	COOLING_DEADBAND , offset from setpoint for cooling to begin. Units are 0.1 deg.
120.0	1.0	HEATING_DEADBAND , offset from setpoint for heating to begin. Units are 0.1 deg.
121.0	1.0	DEGC_OR_F, engineering units, Deg C = 0, Deg F = 1
122.0	1.0	FAN , number of fan speeds. Single speed = 1 up to three speed fan = 3
123.0	1.0	NIGHT_HEATING_DEADBAND , heating deadband in the night time or OFF mode. Units of 1 deg.
124.0	1.0	NIGHT_COOLING_DEADBAND , cooling deadband for the night (OFF) mode. Units of 1 deg.
125.0	1.0	NIGHT HEAT/COOL SETPOINTS , 0 = office (absolute setpoints, 1 = Hotel or Residential, deadband type offsets
126.0	1.0	POWERUP_SETPOINT , setpoint on power up

Address	Bytes	Register and description
127.0	1.0	POWERUP_MODE, mode of operation on power up. 0 = power off, 1 = power up in on mode, 2 = last value (default), 3 = auto mode.
128.0	1.0	KEYPAD_SELECT , variable to select various keypad arrangements. Refer to PAd description in Table 1: Advanced Menu Items
129.0	1.0	AUTO_ONLY , enables or disables manual mode. 0 = Manual Fan Modes 1-x Allowed (depending on R122 value, 1 = Auto Mode Only, 2 = DDC mode,the user can not change setpoint and fan speed from keypad.
130.0	1.0	NOT USED FOR REV 25
131.0	1.0	MAX_SETPOINT, Setpoint high, the highest setpoint a user will be able to set from the keypad.
132.0	1.0	MIN_SETPOINT, Setpoint Low, the lowest setpoint a user will be able to set from the keypad.
133.0	1.0	SPECIAL_MENU_LOCK, Special menu lockout via keypad, serial port only, 0=Full Menu, 1=Menu Disabled, 2=Typical small menu system, 3 = The user can adjust the setpoint in menu mode, 4 = Partial menu enabled When the LOC parameter isset to ‘1’ the keypad will be locked out from all menu operations. The normal operation of the keypad is not affected; the fan and setpoint buttons work as usual. When the LOC parameter is set to ‘2’ the keypad will be locked out from partial menu operations allowing maintenance personnel to access some of the less critical menu parameters while maintaining a LOC on functions reserved for the primary administrator. This option allowsaccess to calibration of the internal and external temperature sensor (CAL and CAE) and the override time parameter (ORT). LOC = 3,The user can not do anything from keypad except enter menu mode.In menu mode,the user can set setpoint ,fan speed,calibration and override timer. When the menu system is locked out, the only way to adjust the tstat parameters is through the network port or through the communications jack at the bottom of the tstat. The parameter can be set back to ‘0’ only though the communications ports as well LOC = 4,User can adjust setpoint and fan speed from keypad,and enter menu to operate part of menu,include CAL,pterm,iterm,SOP,cooling deadband,heating deadband,C/F units and factory default.
134.0	1.0	Write 1 to register 134 - resets the unit to latest factory defaults. (same as writing Writing 143 to register 16)
135.0	1.0	COOLING_SETPOINT, current pid setpoint - limits are set by the max and min setpoints
136.0		Heating setpoint,current heating setpoint - limits are set by the max and min setpoints
137.0	1.0	FAN_SPEED, current operating fan speed

Address	Bytes	Register and description
138.0	1.0	FAN0_OPERATION_TABLE_COAST
139.0	1.0	FAN0_OPERATION_TABLE_COOL1
140.0	1.0	FAN0_OPERATION_TABLE_COOL2
141.0	1.0	FAN0_OPERATION_TABLE_COOL3
142.0	1.0	FAN0_OPERATION_TABLE_HEAT1
143.0	1.0	FAN0_OPERATION_TABLE_HEAT2
144.0	1.0	FAN0_OPERATION_TABLE_HEAT3
145.0	1.0	FAN1_OPERATION_TABLE_COAST
146.0	1.0	FAN1_OPERATION_TABLE_COOL1
147.0	1.0	FAN1_OPERATION_TABLE_COOL2
148.0	1.0	FAN1_OPERATION_TABLE_COOL3
149.0	1.0	FAN1_OPERATION_TABLE_HEAT1
150.0	1.0	FAN1_OPERATION_TABLE_HEAT2
151.0	1.0	FAN1_OPERATION_TABLE_HEAT3
152.0	1.0	FAN2_OPERATION_TABLE_COAST
153.0	1.0	FAN2_OPERATION_TABLE_COOL1
154.0	1.0	FAN2_OPERATION_TABLE_COOL2
155.0	1.0	FAN2_OPERATION_TABLE_COOL3
156.0	1.0	FAN2_OPERATION_TABLE_HEAT1
157.0	1.0	FAN2_OPERATION_TABLE_HEAT2
158.0	1.0	FAN2_OPERATION_TABLE_HEAT3
159.0	1.0	FAN3_OPERATION_TABLE_COAST
160.0	1.0	FAN3_OPERATION_TABLE_COOL1
161.0	1.0	FAN3_OPERATION_TABLE_COOL2
162.0	1.0	FAN3_OPERATION_TABLE_COOL3

Address	Bytes	Register and description
163.0	1.0	FAN3_OPERATION_TABLE_HEAT1
164	1.0	FAN3_OPERATION_TABLE_HEAT2
165.0	1.0	FAN3_OPERATION_TABLE_HEAT3
166.0	1.0	FANAUT_OPERATION_TABLE_COAST
167.0	1.0	FANAUT_OPERATION_TABLE_COOL1
168.0	1.0	FANAUT_OPERATION_TABLE_COOL2
169.0	1.0	FANAUT_OPERATION_TABLE_COOL3
170.0	1.0	FANAUT_OPERATION_TABLE_HEAT1
171.0	1.0	FANAUT_OPERATION_TABLE_HEAT2
172.0	1.0	FANAUT_OPERATION_TABLE_HEAT3
173.0	1.0	VALVE_OPER_TABLE_COAST, Analog output state for each of the 7 modes of operation
174.0	1.0	VALVE_OPER_TABLE_COOLING1
175.0	1.0	VALVE_OPER_TABLE_COOLING2
176.0	1.0	VALVE_OPER_TABLE_COOLING3
177.0	1.0	VALVE_OPER_TABLE_HEATING1
178.0	1.0	VALVE_OPER_TABLE_HEATING2
179.0	1.0	VALVE_OPER_TABLE_HEATING3
180.0	2.0	Analog Input 1, Filtered and calibrated value for analog input 1
181.0	2.0	Analog Input 2, Filtered and calibrated value for analog input 2
182.0	1.0	Night heating setpoint
183.0	1.0	Night cooling setpoint
184.0	1.0	Info Byte, this register contains info about the state of the tstat.
185.0	1.0	Bau - Baudrate, 0=9600, 1=19.2kbaud
186.0	1.0	Ou1 - Output1 Scale - 0=On/Off, 1=0-10V, 2=0-5V, 3=2-10V, 4= 10-0V
187.0	1.0	Ou2 - Output2 Scale - 0=On/Off, 1=0-10V, 2=0-5V, 3=2-10V, 4= 10-0V

Address	Bytes	Register and description
188.0	1.0	AI1 – Analog input 1 range 0=10-bit raw data, 1=10K thermistor, 2=0-100%, 3=on/off, 4=custom
189.0	1.0	AI2 – Analog input 2 range 0=10-bit raw data, 1=10K thermistor, 2=0-100%, 3=on/off, 4=custom
190.0	1.0	dI1 – Digital input 1 range 0 = ON/OFF.
191.0	1.0	OUTPUT1_DELAY_OFF_TO_ON – delay time for output1 going from OFF to ON (sec)
192.0	1.0	OUTPUT2_DELAY_OFF_TO_ON – delay time for output2 going from OFF to ON (sec)
193.0	1.0	OUTPUT3_DELAY_OFF_TO_ON – delay time for output3 going from OFF to ON (sec)
194.0	1.0	OUTPUT4_DELAY_OFF_TO_ON – delay time for output4 going from OFF to ON (sec)
195.0	1.0	OUTPUT5_DELAY_OFF_TO_ON – delay time for output5 going from OFF to ON (sec)
196.0	1.0	OUTPUT1_DELAY_ON_TO_OFF – delay time for output1 going from OFF to ON (sec)
197.0	1.0	OUTPUT2_DELAY_ON_TO_OFF – delay time for output2 going from OFF to ON (sec)
198.0	1.0	OUTPUT3_DELAY_ON_TO_OFF – delay time for output3 going from OFF to ON (sec)
199.0	1.0	OUTPUT4_DELAY_ON_TO_OFF – delay time for output4 going from OFF to ON (sec)
200.0	1.0	OUTPUT5_DELAY_ON_TO_OFF – delay time for output5 going from OFF to ON (sec)
201.0	1.0	MODBUS_CYCLING_DELAY – delay time (in minutes) for switching out of heating or cooling and then back in.
202.0	1.0	MODBUS_CHANGOVER_DELAY – delay time (in minutes) for switching from cooling into heating or vice versa.
203.0	1.0	dIS – Display. This sets the display to either room temperature or setpoint. 0 = room temp, 1 = setpoint, 2 = Blank Display,3 = PID2 value,4 = PID2 setpoint, 5 = set segment code by manually, 6 = Display sleep
204.0	1.0	LED1 (top left to bottom right) — 3.0 — = — RELAY3 — 17.0 — = — FAN MED
205.0	1.0	LED2 — 4.0 — = — RELAY4 — 18.0 — = — FAN HI
206.0	1.0	LED3 — 5.0 — = — RELAY5 — 19.0 — = — FAN AUTO
207.0	1.0	LED4 — 6.0 — = — COASTING STAGE — 20.0 — = — HEAT1 OR COOL1
208.0	1.0	LED5 — 7.0 — = — COOLING1 STAGE — 21.0 — = — HEAT2 OR COOL2

TStat registers, part 2

The Modbus register map for the TStat, in address order. Addresses are decimal, and where the map gives a range the register occupies every address in it. This part begins at address 209.0 and ends at 324.0.

Address	Bytes	Register and description
209.0	1.0	LED6 — 8.0 — = — COOLING2 STAGE — 22.0 — = — HEAT3 OR COOL3
210.0	1.0	LED7 — 9.0 — = — COOLING3 STAGE — 23.0 — = — COOL1, 2, OR 3
10.0	=	HEATING1 STAGE — 24.0 — = — HEAT1, 2, OR 3
11.0	=	HEATING2 STAGE — 25.0 — = — OCCUPANCY
12.0	=	HEATING3 STAGE — 26.0 — = — STAGE 1, 2 OR 3
13.0	=	COOLING MODE — 27.0 — = — STAGE 2 OR 3
211.0	1.0	Unoccupied Override Timer, Ort. 0=disabled, >0=number of minutes manual override is allowed
212.0	1.0	OVERRIDE_TIMER_DOWN_COUNT - Number of minutes remaining on the timer when unoccupied override timer is in effect.
213.0	1.0	Temperature sensor filter, FIL, weighted average of stored value to new raw value
214.0	1.0	Heating cooling mode configuration, HC, 0=PID, 1=Keypad, 2=Digital_in1, 3=Digital_in1, 4=Analog_in1, 5=Analog_in2
215.0	2.0	Internal Temperature Sensor IC - Intenral temp chip, not used in the current hardware
216.0	2.0	Internal Thermistor Sensor - Shows the filtered, calibrated value of the internal thermistor sensor
217.0	2.0	Calibration for the internal thermistor - internally managed offset for the internal temp sensor value
218.0	2.0	Calibration for analog input2 - can write here or write to the analog input and this will be calculated automatically
219.0	2.0	Lookup Table 1 - 0.0V value Sensor value that corresponds to 0.0V
220.0	2.0	Lookup Table 1 - 0.5V value Sensor value that corresponds to 0.5V
221.0	2.0	Lookup Table 1 - 1.0V value Sensor value that corresponds to 1.0V
222.0	2.0	Lookup Table 1 - 1.5V value Sensor value that corresponds to 1.5V
223.0	2.0	Lookup Table 1 - 2.0V value Sensor value that corresponds to 2.0V
224.0	2.0	Lookup Table 1 - 2.5V value Sensor value that corresponds to 2.5V
225.0	2.0	Lookup Table 1 - 3.0V value Sensor value that corresponds to 3.0V
226.0	2.0	Lookup Table 1 - 3.5V value Sensor value that corresponds to 3.5V

Address	Bytes	Register and description
227.0	2.0	Lookup Table 1 - 4.0V value Sensor value that corresponds to 4.0V
228.0	2.0	Lookup Table 1 - 4.5V value Sensor value that corresponds to 4.5V
229.0	2.0	Lookup Table 1 - 5.0V value Sensor value that corresponds to 5.0V
230.0	2.0	Lookup Table 2 - 0.0V value Sensor value that corresponds to 0.0V
231.0	2.0	Lookup Table 2 - 0.5V value Sensor value that corresponds to 0.5V
232.0	2.0	Lookup Table 2 - 1.0V value Sensor value that corresponds to 1.0V
233.0	2.0	Lookup Table 2 - 1.5V value Sensor value that corresponds to 1.5V
234.0	2.0	Lookup Table 2 - 2.0V value Sensor value that corresponds to 2.0V
235.0	2.0	Lookup Table 2 - 2.5V value Sensor value that corresponds to 2.5V
236.0	2.0	Lookup Table 2 - 3.0V value Sensor value that corresponds to 3.0V
237.0	2.0	Lookup Table 2 - 3.5V value Sensor value that corresponds to 3.5V
238.0	2.0	Lookup Table 2 - 4.0V value Sensor value that corresponds to 4.0V
239.0	2.0	Lookup Table 2 - 4.5V value Sensor value that corresponds to 4.5V
240.0	2.0	Lookup Table 2 - 5.0V value Sensor value that corresponds to 5.0V
241.0	2.0	Universal PID input select, 0=none, 1=analog_in1, 2=analog_in2
242.0	2.0	Universal PID upper deadband
243.0	2.0	Universal PID lower deadband
244.0	2.0	Universal PID pterm
245.0	2.0	Universal PID iterm
246.0	2.0	Universal PID setpoint
247.0	1.0	Output 1 PID Control 0 = PID1, can assign each output to either PID1 or 2, the max or the min of the two PIDS
248.0	1.0	Output 2 PID Control 1 = PID2
249.0	1.0	Output 3 PID Control 2 = Maximum of PID1 and PID2
250.0	1.0	Output 4 PID Control 3 = Minimum of PID1 and PID2
251.0	1.0	Output 5 PID Control

Address	Bytes	Register and description
252.0	1.0	Output 6 PID Control
253.0	1.0	Output 7 PID Control
254.0	1.0	PID2 Output table- Coasting
255.0	1.0	PID2 Output table- Cooling1
256.0	1.0	PID2 Output table- Cooling2
257.0	1.0	PID2 Output table- Cooling3
258.0	1.0	PID2 Output - Heating1
259.0	1.0	PID2 Output - Heating
260.0	1.0	PID2 Output - Heating3
261.0	1.0	PID2 Valve Output - Coasting
262.0	1.0	Universal PID Valve Output - Cooling1
263.0	1.0	Universal PID Valve Output - Cooling2
264.0	1.0	Universal PID Valve Output - Cooling3
265.0	1.0	Universal PID Valve Output - Heating1
266.0	1.0	Universal PID Valve Output - Heating2
267.0	1.0	Universal PID Valve Output - Heating3
268.0	1.0	Number of Heating Stages in Universal Table-(Maximum # of total heating and cooling states is 6)
269.0	1.0	Number of Cooling Stages in Universal Table-(Maximum # of total heating and cooling states is 6)
270.0	1.0	PID2, the current value of PID number 2
271.0	2.0	PID1 Units High byte - Upper 2 bytes of the PID1 units in ASCII
272.0	2.0	PID1 Units Low byte - Lower 2 bytes of the PID1 units in ASCII
273.0	2.0	PID2 Units High byte - Upper 2 bytes of the PID2 units in ASCII
274.0	2.0	PID2 Units Low byte - Lower 2 bytes of the PID2 units in ASCII
275.0	2.0	PID2 Unoccupied (OFF) Setpoint

Address	Bytes	Register and description
276.0	1.0	Number of Heating Stages in Original Table - (Maximum # of total heating and cooling states is 6)
277.0	1.0	Number of Cooling Stages in Original Table - (Maximum # of total heating and cooling states is 6)
278.0	1.0	PID2 heating or cooling state derived from PID2 register 240, 0=coasting, 1=cooling1, 2=cooling2, 3=cooling3, 4=heating1, 5=heating2, 6=heating3, 14=cooling4, 15=cooling5, 16=cooling6, 17=heating4, 18=heating5, 19=heating6.
279.0	1.0	Valve travel time. The time of the valve travel from one end to another end. The units is second.
280.0	1.0	Determine the output1 mode. Output1 always is ON/OFF mode
281.0	1.0	Determine the output2 mode. Output2 always is ON/OFF mode
282.0	1.0	Determine the output3 mode. Output3 always is ON/OFF mode
283.0	1.0	Determine the output4 mode. 0, ON/OFF mode; 1, floating valve for cooling; 2, lighting control; 3, PWM
284.0	1.0	Determine the output5 mode. 0, ON/OFF mode; 1, floating valve for heating; 2, lighting control; 3, PWM
285.0	1.0	Valve percent. Show the valve opened how much percent. READ ONLY
286.0	1.0	Interlock for output1
287.0	1.0	Interlock for output2
288.0	1.0	Interlock for output3
289.0	1.0	Interlock for output4
290.0	1.0	Interlock for output5
291.0	1.0	Interlock for output6
292.0	1.0	Interlock for output7
293.0	1.0	Setpoint increment on the display each time the user hits the up/down buttons. Units are 0.1Deg, 10 = 1Deg and so on.
294.0	2.0	Last key pressed counter. Minutes since the used last pressed a key.
295.0	1.0	Freeze protect setpoint. If the ambient temperature less than the setpoint, the heating valve will open some time the Delay to off register set .

Address	Bytes	Register and description
296.0	1.0	Delay to open. The heating valve will open if the ambient temp less than the Freeze temp setpoint last the time this register set. The units is second.
297.0	1.0	Delay to close. The duration the heating valve open. The units is minute.
298.0	1.0	Analog input1 function selection. 0, normal; 1, freeze protect sensor input; 2, occupancy sensor input; 3, sweep off mode; 4, clock mode; 5, change over mode. Refer to dI1 on page13.8 = window interlock
299.0	1.0	Analog input2 function selection. 0, normal; 1, freeze protect sensor input; 2, occupancy sensor input; 3, sweep off mode; 4, clock mode; 5, change over mode. Refer to dI1 on page13.8 =window interlcok
300.0	1.0	dI1 – Digital input 1 function. Refer to dI1 description in Table 1: Advanced Menu Items
301.0	2.0	Period timer ON time.
302.0	2.0	Period timer OFF time.
303.0	1.0	Period timer units. 0, second; 1, minute; 2, hour.
304.0	1.0	Keypad encoded value. Last keypress, READ ONLY
305.0	1.0	LED hundred's digit, , can drive the LEDs manually when the display register 203 is set to manual (5)
306.0	1.0	LED ten's digit code, can drive the LEDs manually when the display register 203 is set to manual (5)
307.0	1.0	LED one's digit code, can drive the LEDs manually when the display register 203 is set to manual (5)
308.0	1.0	LED discrete status symbols, can drive the LEDs manually when the display register 203 is set to manual (5) — Register 309 for TStat5E
309.0	1.0	Input auto/ manual enable. Bit0 correspond to analog input1(register 180); bit1 to analog input2(register 181); bit2 to digital input1(register 311),bit3 correspond to itternal sensor(register 216).0, auto mode, the corresponding input value from sensor; 1, manual mode, the corresponding value from serial port. — Input auto/ manual enable. Bit0 correspond to analog input1(register 367); bit1 to analog input2(register 368); bit7 to analog input8(register 374),bit8 correspond to itternal sensor(register 216).0, auto mode, the corresponding input value
310.0	1.0	Output auto/manual enable. Bit 0 to 4 correspond to output1 to output5, bit 5 correspond to output6(register 102), bit 6 correspond to output7(register 103). 0, auto mode; 1, manual mode. — from sensor; 1, manual mode, the corresponding value from serial port.

Address	Bytes	Register and description
311.0	1.0	Digital manual input. Write the manual value for digital input when digital input in manual mode.
312.0	1.0	(REV35_0) DEADMASTER_AUTO_MANUAL = 0, the default, outputs will not change, stay in whatever mode they were last commanded DEADMASTER_AUTO_MANUAL = 1, the output will be trigger to “AUTO” mode DEADMASTER_AUTO_MANUAL = 2, the outputs will go to manual on or off as defined in register 313.
313.0	1.0	REV 35_0 (relay1-5) output manual value when DEADMASTER_AUTO_MANUAL = 2 register 313, Bit0 is for relay 1, bit1 for relay 2 and so on up to output 5
314.0	2.0	REV 35_0 (analog output - cooling)deadmaster config value during deadmaster be triggered when DEADMASTER_AUTO_MANUAL = 2 Value Range: 0 to 1000
315.0	2.0	REV 35_0 (analog output - heating)deadmaster config value during deadmaster be triggered when DEADMASTER_AUTO_MANUAL = 2 Value Range: 0 to 1000
316.0	1.0	RESERVE
317.0	1.0	Dead master. The Tstat will go to occupied mode automatically if there is no serial communications for a user defined period of time, for example if the register is set to 10 the thermostat will go to ocupied mode if there are no communications for a period of 10 minutes. Set the register to 0 to disable the feature
318.0	1.0	Rouding display. 0 = round the display to the nearest digit; 1 = round the display to the nearest 1/10 unit; 5 = round the display to the nearest 1/2 unit. 2,3,4 reserved.
319.0	1.0	The minimum address which the device can accept, use this to limit addresses to a certain defined band.
320.0	1.0	The maximum device address can be set. The device address should between min and max address, default is 254
321.0	1.0	Rotation features [Old disabled feature] Output 2 is controlled by which output table in the rotation group. READ ONLY.
322.0	1.0	Rotation feature, [disabled]: output3
323.0	1.0	Rotation feature, [disabled]: output4
324.0	1.0	Rotation feature, [disabled]: output5

TStat registers, part 3

The Modbus register map for the TStat, in address order. Addresses are decimal, and where the map gives a range the register occupies every address in it. This part begins at address 325.0 and ends at 444.0.

Address	Bytes	Register and description
325.0	1.0	Rotation time left. Long long time left the rotation will happen. READ ONLY.
326.0	1.0	Show the size of E2 chip. 0 = 24c02, 1 = 24c08/24c16.
327.0	1.0	Assign the timer to be used for which feature. 0 = period timer, 1 = rotation timer, 2 = interlock, 3 = PWM timer.
328.0	1.0	The output1 function, there are three functions for the output1.0 = normal ON/OFF output, 1 = rotation (old disabled feature) 2 = lighting control where one button can be assigned to toggle a relay on & off
329.0	1.0	Rotation Feature: Show which output table is used for output1 when output1 belongs to the rotation group
330.0	1.0	Rotation Feature: Show which output table is used for output1 when output2 belongs to the rotation group
331.0	1.0	Rotation Feature: Show which output table is used for output1 when output3 belongs to the rotation group
332.0	1.0	Rotation Feature: Show which output table is used for output1 when output4 belongs to the rotation group
333.0	2.0	How much time left before output rotation gets retriggered
334.0	1.0	The output2 function setting: 0 = normal ON/OFF output, 1 = rotation, 2 = lighting control.
335.0	1.0	Output3 function setting (see above)
336.0	1.0	Output4 function setting (see above)
337.0	1.0	Output5 function setting (see above)
338.0	1.0	Default occupied setpoint. Works in concert with the "occupied setpoint control register", register 339
339.0	1.0	Occupied Setpoint Control Register: 0 = normal, setpoint is managed by the serial port and keypad, the stat will remember the last occupied setpoint and use that during the next occupied period. 1 = Default mode, the last occupied setpoint if forgotten and the occupied setpoint gets reset to the default. 2 = trigger an event, when a master controller writes 2 to this register, the default setpoint will be copied to the occupied setpoint after which the Tstat will set the value back to 1 to show the event has been serviced.
340.0	1.0	Enable/disable PIR correspond 1/0 respectively.

Address	Bytes	Register and description
341.0	1.0	PWM output range in COAST mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
342.0	1.0	PWM output range in COOLING2 mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
343.0	1.0	PWM output range in COOLING3 mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
344.0	1.0	PWM output range in COOLING1 mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
345.0	1.0	PWM output range in HEATING1 mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
346.0	1.0	PWM output range in HEATING2 mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
347.0	1.0	PWM output range in HEATING3 mode. 0 = CLOSE, 1 = OPEN, 2 = 0-100%, 3 = 50-100%, 4 = 0-50%. MSb 4 bits correspond to output4 and LSB 4 bits correspond to output5
348.0	1.0	The ON period take how many percentage for output4
349.0	1.0	The ON period take how many percentage for output5
350.0	1.0	Free cooling configuration. bit0, free cool enable/disable, 0 = disable, 1= enable. bit1, free cool available decided by local or external master. 0 = local, 1 = NC. bit2, free cool available status, 0 = npt available, 1= available. bit4, show the status if NC is OK when the free cool decided by NC.
351.0	1.0	Analog output OFF table, coasting mode
352.0	1.0	Analog output OFF table, cooling1 mode
353.0	1.0	Analog output OFF table, cooling2 mode
354.0	1.0	Analog output OFF table, cooling3 mode
355.0	1.0	Analog output OFF table, heating1 mode
356.0	1.0	Analog output OFF table, heating2 mode

Address	Bytes	Register and description
357.0	1.0	Analog output OFF table, heating3 mode — The following register for Tstat 5E
358.0	1.0	Register lock. All registers except fan speed and manual inputs/outputs register are not writable. 0 = lock,1 = no lock. — Bytes — Description
359.0	1.0	Outside temperature for free cooling,from external sensor or NC. — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = CUSTOMER, 5 = OFF/ON
360.0	2.0	If outside temp be set from NC. The communication with NC must be set in this time,otherwise will set error status and use external sensor. — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = CUSTOMER, 5 = OFF/ON
361.0	1.0	If the outside air temp is lower than the room temperature by this amount, then the free cooling is worthwhile, 350 bit2 = 1.If the OAT is greater than the room temp, then free cooling mode is not worthwhile. , 350 bit2 = 0 — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = N/A, 5 = OFF/ON
362.0	1.0	Free cooling output configuration. Coasting mode — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = N/A, 5 = OFF/ON
363.0	1.0	Free cooling output configuration.Cooling1 mode — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = N/A, 5 = OFF/ON
364.0	1.0	Free cooling output configuration.Cooling2 mode — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = N/A, 5 = OFF/ON
365.0	1.0	Free cooling output configuration.Cooling3 mode — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = N/A, 5 = OFF/ON
366.0	1.0	Free cooling output configuration.Heating1 mode — 1.0 — ANALOG INPUT RANGE. 0 = raw data, 1 = thermistor, 2 = %, 3 = ON/OFF, 4 = N/A, 5 = OFF/ON
367.0	1.0	Free cooling output configuration.Heating2 mode — 2.0 — Analog input1 value
368.0	1.0	Free cooling output configuration.heating3 mode — 2.0 — Analog input2 value
369.0	1.0	Min Air,the units is percent. Set the minimum output for free cooling, the default is 15% — 2.0 — Analog input3 value
370.0	1.0	Outside air temperature in hottest day — 2.0 — Analog input4 value
371.0	1.0	Outside air temperature in coldest day — 2.0 — Analog input5 value
372.0	1.0	Offset in hottest day — 2.0 — Analog input6 value
373.0	1.0	Offset in coldest day — 2.0 — Analog input7 value

Address	Bytes	Register and description
374.0	2.0	Store setpoint in two bytes, the resolution is 0.1 — 2.0 — Analog input8 value
375.0	1.0	Crrent setpoint = user setpoint + offset setpoint — 2.0 — Calibration for analog input1
376.0	1.0	Setpoint offset — 2.0 — Calibration for analog input2
377.0	1.0	Change over sensor mode, 1 = cooling mode, 0 = heating mode. — 2.0 — Calibration for analog input3
378.0	1.0	external sensor inuse,0 = not use,1 = in use — 2.0 — Calibration for analog input4
379.0	1.0	external sensor status,0 = OK,1 = not OK,damaged or short circuit — 2.0 — Calibration for analog input5
380.0	2.0	day cooling setpoint,work together with heating setpoint in day office mode — 2.0 — Calibration for analog input6
381.0	1.0	daysetpoint option,0 =office,1 = hotel — 2.0 — Calibration for analog input7
382.0	1.0	Timer left for output1 ON to OFF delay — 2.0 — Calibration for analog input8
383.0	1.0	Timer left for output2 ON to OFF delay — 1.0 — The first character on LCD line 1, ASCII code
384.0	1.0	Timer left for output3 ON to OFF delay — 1.0 — The second character on LCD line 1, ASCII code
385.0	1.0	Timer left for output4 ON to OFF delay — 1.0 — The third character on LCD line 1, ASCII code
386.0	1.0	Timer left for output5 ON to OFF delay — 1.0 — The fourth character on LCD line 1, ASCII code
387.0	1.0	Timer left for output1 OFF to ON delay — 1.0 — The fifth character on LCD line 1, ASCII code
388.0	1.0	Timer left for output2 OFF to ON delay — 1.0 — The sixth character on LCD line 1, ASCII code
389.0	1.0	Timer left for output3 OFF to ON delay — 1.0 — The seventh character on LCD line 1, ASCII code
390.0	1.0	Timer left for output4 OFF to ON delay — 1.0 — The eighth character on LCD line 1, ASCII code
391.0	1.0	Timer left for output5 OFF to ON delay — 1.0 — The first character on LCD line 2, ASCII code

Address	Bytes	Register and description
392.0	1.0	The second character on LCD line 2, ASCII code
393.0	1.0	The third character on LCD line 2, ASCII code
394.0	1.0	The fourth character on LCD line 2, ASCII code
395.0	1.0	The fifth character on LCD line 2, ASCII code
396.0	1.0	The sixth character on LCD line 2, ASCII code
397.0	1.0	The seventh character on LCD line 2, ASCII code
398.0	1.0	The eighth character on LCD line 2, ASCII code
399.0	1.0	LCD turn off delay, 0 - 255 minutes
400.0	1.0	Select what parameter will display, 0 = nothing, 1 =temperature, 2 = setpoint, 3 through 10 correspond to input 1 through 8, 11 = mode
401.0	1.0	Select what parameter will display, 0 = nothing, 1 =temperature, 2 = setpoint, 3 through 10 correspond to input 1 through 8, 11 = mode
402.0	1.0	DAY icon control register. 0 = AUTO, 1 = turn OFF, 2 = turn ON.
403.0	1.0	NIGHT icon control register. 0 = AUTO, 1 = turn OFF, 2 = turn ON.
404.0	1.0	Current mode,0 = coasting,1 = heating,2 = cooling
405.0	1.0	Output 1 from ON to OFF time left
406.0	1.0	Output 2 from ON to OFF time left
407.0	1.0	Output 3 from ON to OFF time left
408.0	1.0	Output 4 from ON to OFF time left
409.0	1.0	Output 5 from ON to OFF tilm left
410.0	1.0	Output 1 from OFF to ON time left
411.0	1.0	Output 2 from OFF to ON time left
412.0	1.0	Output 3 from OFF to ON time left
413.0	1.0	Output 4 from OFF to ON time left
414.0	1.0	Output 5 from OFF to ON time left
415.0	1.0	The hottest output air temperature

Address	Bytes	Register and description
416.0	1.0	The coldest output air temperature
417.0	1.0	The offset for the hottest temperature
418.0	1.0	The offset for the coldest temperature
419.0	1.0	Crrent setpoint = user setpoint + offset setpoint
420.0	1.0	The current setpoint offset
421.0	1.0	turn off all fan icons
422.0	2.0	normal cooling setpoint(135) + cooling deadband
423.0	1.0	day setpointoption,1 = hotel,0 = office
424.0	1.0	Window interlock delay,the units is minute, from 0 to 255 minutes
425.0	2.0	User name. line 1:characters 1 & 2
426.0	2.0	User name. line 1:characters 3 & 4
427.0	2.0	User name. line 1:characters 5 & 6 — User name. line 1:characters 5 & 6
428.0	2.0	User name. line 1:characters 7 & 8 — User name. line 1:characters 7 & 8
429.0	2.0	User name. line 2:characters 1 & 2 — User name. line 2:characters 1 & 2
430.0	2.0	User name. line 2:characters 3 & 4 — User name. line 2:characters 3 & 4
431.0	2.0	User name. line 2:characters 5 & 6 — User name. line 2:characters 5 & 6
432.0	2.0	User name. line 2:characters 7 & 8 — User name. line 2:characters 7 & 8
433.0	2.0	Internal sensor name. line 1:characters 1 & 2 — Internal sensor name. line 1:characters 1 & 2 — Internal sensor name. line 1:characters 1 & 2 — Internal sensor name. line 1:characters 1 & 2 — Internal sensor name. line 1:characters 1 & 2 — Internal sensor name. line 1:characters 1 & 2
434.0	2.0	Internal sensor name. line 1:characters 3 & 4 — Internal sensor name. line 1:characters 3 & 4 — Internal sensor name. line 1:characters 3 & 4 — Internal sensor name. line 1:characters 3 & 4 — Internal sensor name. line 1:characters 3 & 4 — Internal sensor name. line 1:characters 3 & 4
435.0	2.0	Internal sensor name. line 1:characters 5 & 6 — Internal sensor name. line 1:characters 5 & 6 — Internal sensor name. line 1:characters 5 & 6 — Internal sensor name. line 1:characters 5 & 6 — Internal sensor name. line 1:characters 5 & 6 — Internal sensor name. line 1:characters 5 & 6

Address	Bytes	Register and description
436.0	2.0	Internal sensor name. line 1:characters 7 & 8 — Internal sensor name. line 1:characters 7 & 8 — Internal sensor name. line 1:characters 7 & 8 — Internal sensor name. line 1:characters 7 & 8 — Internal sensor name. line 1:characters 7 & 8 — Internal sensor name. line 1:characters 7 & 8
437.0	2.0	Input 1 user name. line 1:characters 1 & 2 — Input 1 user name. line 1:characters 1 & 2 — Input 1 user name. line 1:characters 1 & 2 — Input 1 user name. line 1:characters 1 & 2 — Input 1 user name. line 1:characters 1 & 2 — Input 1 user name. line 1:characters 1 & 2
438.0	2.0	Input 1 user name. line 1:characters 3 & 4 — Input 1 user name. line 1:characters 3 & 4 — Input 1 user name. line 1:characters 3 & 4 — Input 1 user name. line 1:characters 3 & 4 — Input 1 user name. line 1:characters 3 & 4 — Input 1 user name. line 1:characters 3 & 4
439.0	2.0	Input 1 user name. line 1:characters 5 & 6 — Input 1 user name. line 1:characters 5 & 6 — Input 1 user name. line 1:characters 5 & 6 — Input 1 user name. line 1:characters 5 & 6 — Input 1 user name. line 1:characters 5 & 6 — Input 1 user name. line 1:characters 5 & 6
440.0	2.0	Input 1 user name. line 1:characters 7 & 8 — Input 1 user name. line 1:characters 7 & 8 — Input 1 user name. line 1:characters 7 & 8 — Input 1 user name. line 1:characters 7 & 8 — Input 1 user name. line 1:characters 7 & 8 — Input 1 user name. line 1:characters 7 & 8
441.0	2.0	Input 2 user name. line 1:characters 1 & 2 — Input 2 user name. line 1:characters 1 & 2 — Input 2 user name. line 1:characters 1 & 2 — Input 2 user name. line 1:characters 1 & 2 — Input 2 user name. line 1:characters 1 & 2 — Input 2 user name. line 1:characters 1 & 2
442.0	2.0	Input 2 user name. line 1:characters 3 & 4 — Input 2 user name. line 1:characters 3 & 4 — Input 2 user name. line 1:characters 3 & 4 — Input 2 user name. line 1:characters 3 & 4 — Input 2 user name. line 1:characters 3 & 4 — Input 2 user name. line 1:characters 3 & 4
443.0	2.0	Input 2 user name. line 1:characters 5 & 6 — Input 2 user name. line 1:characters 5 & 6 — Input 2 user name. line 1:characters 5 & 6 — Input 2 user name. line 1:characters 5 & 6 — Input 2 user name. line 1:characters 5 & 6 — Input 2 user name. line 1:characters 5 & 6
444.0	2.0	Input 2 user name. line 1:characters 7 & 8 — Input 2 user name. line 1:characters 7 & 8 — Input 2 user name. line 1:characters 7 & 8 — Input 2 user name. line 1:characters 7 & 8 — Input 2 user name. line 1:characters 7 & 8 — Input 2 user name. line 1:characters 7 & 8

TStat registers, part 4

The Modbus register map for the TStat, in address order. Addresses are decimal, and where the map gives a range the register occupies every address in it. This part begins at address 445.0 and ends at 469.0.

Address	Bytes	Register and description
445.0	2.0	Input 3 user name. line 1:characters 1 & 2 — Input 3 user name. line 1:characters 1 & 2 — Input 3 user name. line 1:characters 1 & 2 — Input 3 user name. line 1:characters 1 & 2 — Input 3 user name. line 1:characters 1 & 2 — Input 3 user name. line 1:characters 1 & 2
446.0	2.0	Input 3 user name. line 1:characters 3 & 4 — Input 3 user name. line 1:characters 3 & 4 — Input 3 user name. line 1:characters 3 & 4 — Input 3 user name. line 1:characters 3 & 4 — Input 3 user name. line 1:characters 3 & 4 — Input 3 user name. line 1:characters 3 & 4
447.0	2.0	Input 3 user name. line 1:characters 5 & 6 — Input 3 user name. line 1:characters 5 & 6 — Input 3 user name. line 1:characters 5 & 6 — Input 3 user name. line 1:characters 5 & 6 — Input 3 user name. line 1:characters 5 & 6 — Input 3 user name. line 1:characters 5 & 6
448.0	2.0	Input 3 user name. line 1:characters 7 & 8 — Input 3 user name. line 1:characters 7 & 8 — Input 3 user name. line 1:characters 7 & 8 — Input 3 user name. line 1:characters 7 & 8 — Input 3 user name. line 1:characters 7 & 8 — Input 3 user name. line 1:characters 7 & 8
449.0	2.0	Input 4 user name. line 1:characters 1 & 2 — Input 4 user name. line 1:characters 1 & 2 — Input 4 user name. line 1:characters 1 & 2 — Input 4 user name. line 1:characters 1 & 2 — Input 4 user name. line 1:characters 1 & 2 — Input 4 user name. line 1:characters 1 & 2
450.0	2.0	Input 4 user name. line 1:characters 3 & 4 — Input 4 user name. line 1:characters 3 & 4 — Input 4 user name. line 1:characters 3 & 4 — Input 4 user name. line 1:characters 3 & 4 — Input 4 user name. line 1:characters 3 & 4 — Input 4 user name. line 1:characters 3 & 4
451.0	2.0	Input 4 user name. line 1:characters 5 & 6 — Input 4 user name. line 1:characters 5 & 6 — Input 4 user name. line 1:characters 5 & 6 — Input 4 user name. line 1:characters 5 & 6 — Input 4 user name. line 1:characters 5 & 6 — Input 4 user name. line 1:characters 5 & 6
452.0	2.0	Input 4 user name. line 1:characters 7 & 8 — Input 4 user name. line 1:characters 7 & 8 — Input 4 user name. line 1:characters 7 & 8 — Input 4 user name. line 1:characters 7 & 8 — Input 4 user name. line 1:characters 7 & 8 — Input 4 user name. line 1:characters 7 & 8
453.0	2.0	Input 5 user name. line 1:characters 1 & 2 — Input 5 user name. line 1:characters 1 & 2 — Input 5 user name. line 1:characters 1 & 2 — Input 5 user name. line 1:characters 1 & 2 — Input 5 user name. line 1:characters 1 & 2 — Input 5 user name. line 1:characters 1 & 2

Address	Bytes	Register and description
454.0	2.0	Input 5 user name. line 1:characters 3 & 4 — Input 5 user name. line 1:characters 3 & 4 — Input 5 user name. line 1:characters 3 & 4 — Input 5 user name. line 1:characters 3 & 4 — Input 5 user name. line 1:characters 3 & 4 — Input 5 user name. line 1:characters 3 & 4
455.0	2.0	Input 5 user name. line 1:characters 5 & 6 — Input 5 user name. line 1:characters 5 & 6 — Input 5 user name. line 1:characters 5 & 6 — Input 5 user name. line 1:characters 5 & 6 — Input 5 user name. line 1:characters 5 & 6 — Input 5 user name. line 1:characters 5 & 6
456.0	2.0	Input 5 user name. line 1:characters 7 & 8 — Input 5 user name. line 1:characters 7 & 8 — Input 5 user name. line 1:characters 7 & 8 — Input 5 user name. line 1:characters 7 & 8 — Input 5 user name. line 1:characters 7 & 8 — Input 5 user name. line 1:characters 7 & 8
457.0	2.0	Input 6 user name. line 1:characters 1 & 2 — Input 6 user name. line 1:characters 1 & 2 — Input 6 user name. line 1:characters 1 & 2 — Input 6 user name. line 1:characters 1 & 2 — Input 6 user name. line 1:characters 1 & 2 — Input 6 user name. line 1:characters 1 & 2
458.0	2.0	Input 6 user name. line 1:characters 3 & 4 — Input 6 user name. line 1:characters 3 & 4 — Input 6 user name. line 1:characters 3 & 4 — Input 6 user name. line 1:characters 3 & 4 — Input 6 user name. line 1:characters 3 & 4 — Input 6 user name. line 1:characters 3 & 4
459.0	2.0	Input 6 user name. line 1:characters 5 & 6 — Input 6 user name. line 1:characters 5 & 6 — Input 6 user name. line 1:characters 5 & 6 — Input 6 user name. line 1:characters 5 & 6 — Input 6 user name. line 1:characters 5 & 6 — Input 6 user name. line 1:characters 5 & 6
460.0	2.0	Input 6 user name. line 1:characters 7 & 8 — Input 6 user name. line 1:characters 7 & 8 — Input 6 user name. line 1:characters 7 & 8 — Input 6 user name. line 1:characters 7 & 8 — Input 6 user name. line 1:characters 7 & 8 — Input 6 user name. line 1:characters 7 & 8
461.0	2.0	Input 7 user name. line 1:characters 1 & 2 — Input 7 user name. line 1:characters 1 & 2 — Input 7 user name. line 1:characters 1 & 2 — Input 7 user name. line 1:characters 1 & 2 — Input 7 user name. line 1:characters 1 & 2 — Input 7 user name. line 1:characters 1 & 2
462.0	2.0	Input 7 user name. line 1:characters 3 & 4 — Input 7 user name. line 1:characters 3 & 4 — Input 7 user name. line 1:characters 3 & 4 — Input 7 user name. line 1:characters 3 & 4 — Input 7 user name. line 1:characters 3 & 4 — Input 7 user name. line 1:characters 3 & 4

Address	Bytes	Register and description
463.0	2.0	Input 7 user name. line 1:characters 5 & 6 — Input 7 user name. line 1:characters 5 & 6 — Input 7 user name. line 1:characters 5 & 6 — Input 7 user name. line 1:characters 5 & 6 — Input 7 user name. line 1:characters 5 & 6 — Input 7 user name. line 1:characters 5 & 6
464.0	2.0	Input 7 user name. line 1:characters 7 & 8 — Input 7 user name. line 1:characters 7 & 8 — Input 7 user name. line 1:characters 7 & 8 — Input 7 user name. line 1:characters 7 & 8 — Input 7 user name. line 1:characters 7 & 8 — Input 7 user name. line 1:characters 7 & 8
465.0	2.0	Input 8 user name. line 1:characters 1 & 2 — Input 8 user name. line 1:characters 1 & 2 — Input 8 user name. line 1:characters 1 & 2 — Input 8 user name. line 1:characters 1 & 2 — Input 8 user name. line 1:characters 1 & 2 — Input 8 user name. line 1:characters 1 & 2
466.0	2.0	Input 8 user name. line 1:characters 3 & 4 — Input 8 user name. line 1:characters 3 & 4 — Input 8 user name. line 1:characters 3 & 4 — Input 8 user name. line 1:characters 3 & 4 — Input 8 user name. line 1:characters 3 & 4 — Input 8 user name. line 1:characters 3 & 4
467.0	2.0	Input 8 user name. line 1:characters 5 & 6 — Input 8 user name. line 1:characters 5 & 6 — Input 8 user name. line 1:characters 5 & 6 — Input 8 user name. line 1:characters 5 & 6 — Input 8 user name. line 1:characters 5 & 6 — Input 8 user name. line 1:characters 5 & 6
468.0	2.0	Input 8 user name. line 1:characters 7 & 8 — Input 8 user name. line 1:characters 7 & 8 — Input 8 user name. line 1:characters 7 & 8 — Input 8 user name. line 1:characters 7 & 8 — Input 8 user name. line 1:characters 7 & 8 — Input 8 user name. line 1:characters 7 & 8
469.0	1.0	lcd rotate enable, decide how many items will be shown on rotate mode,range0 to 12

Register notes and model tables

The end of the register workbook is prose rather than a map: notes that apply to the registers above, and the table of TStat5 models with what each one displays. It is kept in the order it appears, because the notes refer to each other by register number.

- Number of buttons on the keypad The keypad can have up to six buttons. The setting is not normally adjusted in the field. Care should be taken to coordinate with the settings in register 106, the Heat / Cool changeover parameter 128=0 , two buttons, for adjusting the setpoint. 128=1 , 4 buttons, lower pair for the mode and upper pair for the setpoint. 128=2 , 6 button keypad, with heat cool manual selection. Lower pair for the mode, next pair for the setpoint and upper pair for the heat or cool mode. 128=3 , 6 button keypad, with separate heating and cooling setpoints. Lower pair for the mode, next pair for the cooling setpoint and uppermost pair for the heating setpoint.
- Relay Output Tables (bit0 = relay1, bit1 = relay2, bit2 = relay3, bit3 = relay4, bit4 = relay5)
- Fan0 table is for the off state. Fan1, Fan2, and Fan3 are for the manual states. Fan4 is for the Auto state. These states are controlled by the user.
- The mode of operation (coasting, cooling, heating) is determined by the PID parameter.
- Analog output OFF table, coasting mode
- Bit 0 is read/write and shows the occupancy mode. Bit 0 = 0 means unoccupied. Bit 0 = 1 means occupied.
- Bit 1 is read only and shows the reset state. Bit 1 = 0 means hardware restart. Bit 1 = 1 means software restart.
- Bit 2 is read/write and is the reset prevention bit. Bit 2 = 0 means the tstat will automatically reset after certain registers are changed. Bit 2 = 1 prevents this reset. Changing this bit from 1 to 0 will trigger a reset.
- Bit 3 is the state of the digital input. Bit 3 = 1 means logic high. Bit 3 = 0 means logic low.
- Bit 4: Reserved, used for some non standard occupancy sensor logic
- Bit5 0=no delay on modbus reply, 1= 10ms delay before send for slower PLC's to switch from TX to RX
- Bit6 0=ID can change, 1= ID can not change
- Bit7, RS485/wireless communications mode: The normal communications method is a bus topology using RS485 which uses a ‘transmit enable’ or TX_EN line on the RS485 hardware whenever transmission from the thermostat to the bus takes place. For wireless devices this is typically taken care of by the radio module itself so it is not needed. Default = 0, When bit7 is 0, the RS485 chip,

TX_EN line is used for normal RS485 bus communications. When bit7 is 1, the TX_EN line is not used, allowing the radio module to communicate one-to-one with the Tstat

- LED TABLE — 0.0 — = — NONE — 14.0 — = — HEATING MODE
- Determines what activates — 1.0 — = — RELAY1 — 15.0 — = — FAN OFF
- the LEDs — 2.0 — = — RELAY2 — 16.0 — = — FAN LOW
- Analog Output Tables (bit0,1=analog out1, bit2,3=analog out2, 00=0%, 01=0-100%, 11=100%)
- Interlock for each output, analog and digital output. 0, interlock always ON; 1, DI1 determine the interlock status ; 2, AI1 determine the interlock status, the range of AI1 must be ON/OFF; 3, AI2 determine the interlock status, the range of AI2 must be ON/OFF; 4, TIMER OR, the output OR with the period timer; 5, TIMER AND, the output AND with the period timer.
- Analog Output Tables (bit0,1 =analog out1, bit2,3=analog out2, 00=0%, 01=0-100%, 11=100%)
- Output table in free cooling mode,0 = 0%, 1 = 100%,2 = MIN->100%,3 = MIN|100%,4 = MIN.Bit7 through 4 correspond to OFF table,bit 3 through 0 correspond to ON table.
- Register list for Tstat5 series Thermostat
- Model — Model Number — Display — AI — DI — AO — DO — DO hardware
- Tstat5A — 2.0 — LED — 1.0 — 0.0 — 2.0 — 3.0 — 3 big relays
- Tstat5B — 1.0 — LED — 1.0 — 0.0 — 0.0 — 5.0 — 2 small + 3 big relays
- Tstat5C — 4.0 — LED — 2.0 — 1.0 — 0.0 — 5.0 — 5 small relays
- Tstat5D — 12.0 — LED — 2.0 — 1.0 — 2.0 — 5.0 — 5 small relays
- Tstat5E — 16.0 — LCD — 8.0 — 0.0 — 2.0 — 5.0 — 5 small relays
- Tstat5F — 17.0 — LED — 2.0 — 0.0 — 0.0 — 5.0 — 2 triac + 3 small relays
- Tstat5G — 18.0 — LED — 2.0 — 0.0 — 2.0 — 5.0 — 2 triac + 3 small relays
- Tstat5H — 19.0 — LCD — 8.0 — 0.0 — 7.0 — 0.0 — 0.0
- Inputs/Outputs register for Tstat5 series Thermostat
- 5A — 5B — 5C — 5D — 5E — 5F — 5G — 5H
- OUT1 — 108.0 — 108.0 — 108.0 — 108.0 — 108.0 — 108.0 — 108.0 — 102.0
- OUT2 — 108.1 — 108.1 — 108.1 — 108.1 — 108.1 — 108.1 — 108.1 — 103.0
- OUT3 — 108.2 — 108.2 — 108.2 — 108.2 — 108.2 — 108.2 — 108.2 — 387.0

- OUT4 — 108.3 — 108.3 — 108.3 — 108.3 — 348.0 — 348.0 — 388.0
- OUT5 — 108.4 — 108.4 — 108.4 — 108.4 — 349.0 — 349.0 — 389.0
- OUT6 — 102.0 — 102.0 — 102.0 — 102.0 — 390.0
- OUT7 — 103.0 — 103.0 — 103.0 — 103.0 — 391.0
- IN1 — 101.0 — 101.0 — 101.0 — 101.0 — 367.0 — 101.0 — 101.0 — 349.0
- IN2 — 180.0 — 180.0 — 180.0 — 180.0 — 368.0 — 180.0 — 180.0 — 350.0
- IN3 — 181.0 — 181.0 — 369.0 — 181.0 — 181.0 — 351.0
- IN4 — 311.0 — 311.0 — 370.0 — 311.0 — 311.0 — 352.0
- IN5 — 371.0 — 353.0
- IN6 — 372.0 — 354.0
- IN7 — 373.0 — 355.0
- IN8 — 374.0 — 356.0
- Inputs/Outputs Range Setting Registers
- 5A — 5B — 5C — 5D — 5E — 5F — 5G — 5H
- OUT1 — 280.0 — 280.0 — 280.0 — 280.0 — 280.0 — 280.0 — 280.0
- OUT2 — 281.0 — 281.0 — 281.0 — 281.0 — 281.0 — 281.0 — 281.0
- OUT3 — 282.0 — 282.0 — 282.0 — 282.0 — 282.0 — 282.0 — 282.0
- OUT4 — 283.0 — 283.0 — 283.0 — 283.0 — 283.0 — 283.0
- OUT5 — 284.0 — 284.0 — 284.0 — 284.0 — 284.0 — 284.0
- OUT6 — 186.0 — 186.0 — 186.0 — 186.0
- OUT7 — 187.0 — 187.0 — 187.0 — 187.0
- IN1 — 121.0 — 121.0 — 121.0 — 121.0 — 359.0 — 121.0 — 121.0 — 341.0
- IN2 — 188.0 — 188.0 — 188.0 — 188.0 — 360.0 — 188.0 — 188.0 — 342.0
- IN3 — 189.0 — 189.0 — 361.0 — 189.0 — 189.0 — 343.0
- IN4 — 190.0 — 190.0 — 362.0 — 190.0 — 190.0 — 344.0
- IN5 — 363.0 — 345.0

- IN6 — 364.0 — 346.0
- IN7 — 365.0 — 347.0
- IN8 — 366.0 — 348.0
- Inputs/Outputs AUTO/MANUAL Registers
- 5A — 5B — 5C — 5D — 5E — 5F — 5G — 5H
- OUT1 — 310.0 — 310.0 — 310.0 — 310.0 — 310.0 — 310.0 — 310.0
- OUT2 — 310.1 — 310.1 — 310.1 — 310.1 — 310.1 — 310.1 — 310.1
- OUT3 — 310.2 — 310.2 — 310.2 — 310.2 — 310.2 — 310.2 — 310.2
- OUT4 — 310.3 — 310.3 — 310.3 — 310.3 — 310.3 — 310.3
- OUT5 — 310.4 — 310.4 — 310.4 — 310.4 — 310.4 — 310.4
- OUT6 — 310.5 — 310.5 — 310.5 — 310.5
- OUT7 — 310.6 — 310.6 — 310.6 — 310.6
- IN1 — 309.0 — 309.0 — 309.0 — 309.0 — 309.0 — 309.0 — 309.0
- IN2 — 309.1 — 309.1 — 309.1 — 309.1 — 309.1 — 309.1 — 309.1
- IN3 — 309.2 — 309.2 — 309.2 — 309.2 — 309.2
- IN4
- IN5
- IN6
- IN7
- IN8

BACnet points by controller

What each T3 model exposes as BACnet objects, and how many of each. The object names are the ones the controller publishes, so they can be matched against a point list read from the device.

T3-BB

Item	Value	Notes
Controller	T3-BB Big Brother	Range
Total Channels	56	
Universal Input	32	
ANALOG_INPUT1 (IN1)	Select "Analog Units" or "No Units"for the corresponding Inputs	ANALOG_INPUT32 (IN32)
BINARY_INPUT1 (IN1)	Select "Digital Units" for the corresponding Inputs	BINARY_INPUT32 (IN32)
Relay Output	12	
BINARY_OUTPUT1 (OUT1)	Select "Digital Units"for the corresponding Outputs	BINARY_OUTPUT12 (OUT12)
Analog Output	12	
ANALOG_OUTPUT13 (OUT13)	Select "Analog Units"for the corresponding Outputs	ANALOG_OUTPUT24 (OUT24)
Variable	128	
ANALOG_VALUE1 VAR1	Select "Analog Units"for the corresponding VAR	ANALOG_VALUE128 VAR128
BINARY_VALUE1 VAR1	Select "Digital Units"for the corresponding VAR	BINARY_VALUE128 VAR128

T3-LB

Item	Value	Notes
Controller	T3-LB Little Brother	Range
Total Channels	26	
Universal Input	16	
ANALOG_INPUT1 (IN1)	Select "Analog Units" or "No Units"for the corresponding Inputs	ANALOG_INPUT16 (IN16)
BINARY_INPUT1 (IN1)	Select "Digital Units" for the corresponding Inputs	BINARY_INPUT16 (IN16)
Relay Output	6	
BINARY_OUTPUT1 (OUT1)	Select "Digital Units"for the corresponding Outputs	BINARY_OUTPUT6 (OUT6)
Analog Output	4	
ANALOG_OUTPUT7 (OUT7)	Select "Analog Units"for the corresponding Outputs	ANALOG_OUTPUT10 (OUT10)
Variable	128	
ANALOG_VALUE1 VAR1	Select "Analog Units"for the corresponding VAR	ANALOG_VALUE128 VAR128
BINARY_VALUE1 VAR1	Select "Digital Units"for the corresponding VAR	BINARY_VALUE128 VAR128

T3-TB

Item	Value	Notes
Controller	T3-TB Tiny Brother	Range
Total Channels	22	
Universal Input	8	
ANALOG_INPUT1 (IN1)	Select "Analog Units" or "No Units"for the corresponding Inputs	ANALOG_INPUT32 (IN32)
BINARY_INPUT1 (IN1)	Select "Digital Units" for the corresponding Inputs	BINARY_INPUT32 (IN32)
Relay Output	8	
BINARY_OUTPUT1 (OUT1)	Select "Digital Units"for the corresponding Outputs	BINARY_OUTPUT8 (OUT8)
Analog Output	6	
ANALOG_OUTPUT9 (OUT9)	Select "Analog Units"for the corresponding Outputs	ANALOG_OUTPUT14 (OUT14)
Variable	128	
ANALOG_VALUE1 VAR1	Select "Analog Units"for the corresponding VAR	ANALOG_VALUE128 VAR128
BINARY_VALUE1 VAR1	Select "Digital Units"for the corresponding VAR	BINARY_VALUE128 VAR128

WireGuard Modbus Register Map

Overview

This document describes the Modbus register interface used to configure WireGuard VPN functionality on T3 ESP32 controllers.

WireGuard configuration is exposed through Modbus Holding Registers and can be accessed using:

- Function Code 03 (Read Holding Registers)
- Function Code 06 (Write Single Register)
- Function Code 16 (Write Multiple Registers)

All configuration changes are automatically saved to flash memory.

Register Allocation

Register Range	Description
2100	WireGuard Enable
2101 - 2133	Private Key
2134 - 2166	Peer Public Key
2167 - 2199	Pre-Shared Key
2200 - 2203	Local VPN IP Address
2204	WireGuard Port
2205 - 2208	Peer Endpoint IP Address
2209 - 2250	Reserved

Register Summary

Register	Name	Access	Description
2100	WireGuard Enable	R/W	Enable or disable WireGuard
2101-2133	Private Key	R/W	Local WireGuard private key
2134-2166	Peer Public Key	R/W	Remote peer public key
2167-2199	Pre-Shared Key	R/W	Optional pre-shared key
2200	Local IP Octet 1	R/W	Local VPN address
2201	Local IP Octet 2	R/W	Local VPN address
2202	Local IP Octet 3	R/W	Local VPN address
2203	Local IP Octet 4	R/W	Local VPN address
2204	WireGuard Port	R/W	UDP port
2205	Peer IP Octet 1	R/W	Peer endpoint address
2206	Peer IP Octet 2	R/W	Peer endpoint address
2207	Peer IP Octet 3	R/W	Peer endpoint address
2208	Peer IP Octet 4	R/W	Peer endpoint address
2209-2250	Reserved	N/A	Reserved for future use

WireGuard Enable

Register 2100

Controls whether the WireGuard interface is enabled.

Value	Description
0	Disabled
1	Enabled

Example:

```
2100 = 1
```

WireGuard is enabled.

Key Storage Format

The following register blocks use identical encoding:

- Private Key (2101-2133)
- Peer Public Key (2134-2166)
- Pre-Shared Key (2167-2199)

Each Modbus register stores **two ASCII characters**.

Register Encoding

```
High Byte = Character 1
Low Byte  = Character 2
```

Example:

```
Characters: A B

'A' = 0x41
'B' = 0x42

Register Value = 0x4142
```

Reading register value:

```
0x4142
```

Results in:

```text
Character 1 = 'A'
Character 2 = 'B'
```

Storage Capacity

Each key block contains:

```text
33 Registers × 2 Bytes = 66 Bytes
```

This capacity is sufficient for standard WireGuard Base64 encoded keys.

Private Key

Registers 2101 - 2133

Stores the local WireGuard private key.

Property	Value
Access	R/W
Capacity	66 Bytes
Encoding	Two ASCII characters per register

Peer Public Key

Registers 2134 - 2166

Stores the remote WireGuard peer public key.

Property	Value
Access	R/W
Capacity	66 Bytes
Encoding	Two ASCII characters per register

Pre-Shared Key

Registers 2167 - 2199

Stores the optional WireGuard pre-shared key.

Property	Value
Access	R/W
Capacity	66 Bytes
Encoding	Two ASCII characters per register

Local VPN Address

Registers 2200 - 2203

The VPN address is configured using four individual octets.

Example:

```text
10.0.0.2
```

Register	Value
2200	10
2201	0
2202	0
2203	2

WireGuard Port

Register 2204

Specifies the UDP port used by WireGuard.

Example:

```
2204 = 51820
```

Supported Range:

```
1 - 65535
```

The same port is used for:

- Local listening
- Peer communication

Peer Endpoint Address

Registers 2205 - 2208

Specifies the remote peer endpoint IP address.

Example:

```
203.0.113.10
```

Register	Value
2205	203
2206	0
2207	113
2208	10

Fixed Parameters

The following WireGuard parameters are fixed within firmware and are not configurable through Modbus.

VPN Netmask

```
255.255.255.0
```

Persistent Keepalive

```
25 Seconds
```

Ping Address

```
10.0.0.1
```

Configuration Persistence

After a successful write to any WireGuard register, the controller automatically saves the updated configuration to non-volatile storage.

No additional save command is required.

Multiple Register Write Support

WireGuard key blocks support Modbus Function Code 16 (Write Multiple Registers).

Example:

```
Start Address : 2101
Quantity      : 33 Registers
```

This is the recommended method for writing:

- Private Key
- Peer Public Key
- Pre-Shared Key

The firmware automatically decodes the register values and stores the resulting key data.

Example Configuration

Local Device

```
TEXT

WireGuard Enable : 1
Local VPN IP     : 10.0.0.2
Port             : 51820
Peer Endpoint    : 203.0.113.10
```

Register Values

Register	Value
2100	1
2200	10
2201	0
2202	0
2203	2
2204	51820
2205	203
2206	0
2207	113
2208	10

Future Expansion

Registers `2209 - 2250` are reserved for future WireGuard enhancements such as:

- Tunnel Status
- Connection State
- Last Handshake Time
- RX/TX Statistics
- Peer Monitoring

- Diagnostic Information
- Multi-Peer Support

Future firmware versions may utilize these registers while maintaining backward compatibility.